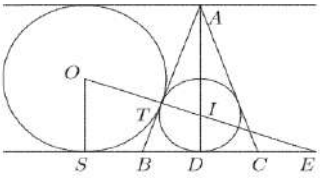


From $1-(1-x)-y = 1-KB-LB = KL = \sqrt{(1-x)^2+y^2}$, we have $x^2-2xy+y^2 = 1-2x+x^2+y^2$, which simplifies to $2x-2xy = 1 \Longrightarrow 2x(1-y) = 1$. Hence the area of triangle DMN is equal to $\frac{1}{2}DM \times DN = \frac{1}{2}x(1-y) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

(b) Let O be the centre of the larger circle and let this circle be tangent to BC at S. Let T be the point of common tangency of the two circles, and let I be the centre of the circle inscribing triangle ABC. Since $AB = AC$, I lies on the altitude AD and D is also the point of tangency of the incircle with BC.



Extend OI to meet BC at E. Let the inradius of ABC be r. Since $OS = 1$, then $AD = 2$. Triangles BAD and BET are congruent since $< ABD =< EBT$.

Now $< ADB = 90^\circ =< ETB$ and $BD = BT$. Hence $ET = 2$ and $EI = 2-r$. Now triangles SOE and DIE are similar.

Hence $r = \frac{ID}{OS} = \frac{IE}{OE} = \frac{2-r}{3}$. It follows that $3r = 2-r \Longrightarrow r = \frac{1}{2}$.

NMC 2015 A-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. Find the positive value of M for which $22222^2 \times 55555^2 = 10^2 \times M^2$?

Qn2. While Annet was reading a novel of less than 1000 pages, she noticed that the sum of the digits on the page she was reading was 19 and the sum of the digits on the next page was 2. What is the number of the page that Annet was reading?

Qn3. The sum of the two 5-digit numbers UMS14 and UMS15 is 123429. What is the value of U+M+S?

Qn4. Nineteen students, numbered 1 through 19 stood in a circle in a clockwise numerical order, all facing the centre. They began to count out loud in a clockwise order: The first called out the number 1, the second called out 2 and each student then called out the number 1 more than the number called to his right. What was the number of the student who called out 2015?

Qn5. Whenever brothers A, B and C are asked a question, TWO lie and ONE tells the truth. I asked who amongst them was the youngest. A said that he was not the oldest. B replied that C was the oldest while C said that A was the youngest. Who actually was the youngest among the three brothers?

Qn6. All but two of my math books are titled *Algebra*, all but two are titled *Trigonometry* and all but two are titled *Geometry*. I have atleast one math book titled *Algebra*. How many math books do I have?

Qn7. Let $F = \log\left(\frac{1+y}{1-y}\right)$. When each y in F is replaced by the expression $\frac{3y+y^3}{1+3y^2}$, a new function H is formed. Find H in terms of F.

Qn8. Alex and Barbra both tried to solve the equation, $Ax^2 + Bx + C = 0$. However, each made exactly one error in copying the equation. The solutions of Alex's equation were -2 and $\frac{-3}{2}$ but his value of A was wrong. The solutions of Barbra's equation were 2 and 3, but her value of B was wrong. What are the solutions of the correct equation?

Qn9. Let $m + n = 1$ where $m = \log_{y^2} x$ and $n = \log_{x^2} y$, and $y > 0, y \neq 1, x \neq 1$. Find the value of x in terms of y.

Qn10. Johnson rides his bike 3 kilometres each day to school. The first kilometre is uphill and he rides at a constant speed of 4 km/hr. The second kilometre is downhill, where he moves at a constant speed that is an integer and is faster than his uphill speed and is less than 60 km/hr. For the final distance, he rides at a constant speed of 6 km/hr. If his average speed, in km/hr, for the entire trip to school is an integer, find the speed, in km/hr, that he moves with down the hill.

SECTION B

Qn11. The geometric sequence with n terms $t_1, t_2, \ldots, t_{n-1}, t_n$ has $t_1 \times t_n = 3$. Also, the product of all n terms equals 59049 (that is, $t_1 \times t_2 \times \ldots \times t_{n-1} \times t_n = 59049$). Determine the value of n.

Qn12. A telephone number has the form ABC-DEF-GHIJ, where each letter represents a different digit. The digits in each part of the number are in decreasing order; that is, $A > B > C, D > E > F$, and $G > H > I > J$. Furthermore, D, E, and F are consecutive even digits; G, H, I, and J are consecutive odd digits; and $A + B + C = 9$. What is the telephone number?

Qn13. When the mean, median, and mode of $10, 2, 5, 2, 4, 2, x$ are arranged in ascending order, they form a non-constant arithmetic progression. What is the sum of all possible real values of x?

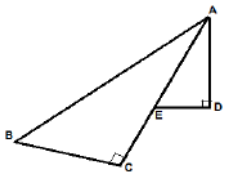
Qn14. The square below is a multiplicative magic square, that is, the product of the numbers in each row, column and diagonal is the same. If all the entries are positive integers, what is the sum of the possible values of g? [Hint: Express each unknown in terms of b].

50	b	c
d	e	f
g	h	2

Qn15. In the diagram below, $< ACB =< ADE = 90°$. If $\overline{AB} = 75$cm, $\overline{BC} = 21$cm, $\overline{AD} = 20$cm and $\overline{CE} = 47$cm. Determine the length of BD.

NMC 2015 A-LEVEL PAPER 2 SOLUTIONS

SECTION A



Qn1.

$$\begin{aligned} 22222^2 \times 55555^2 &= 10^2 \times M^2 \\ (2 \times 11111)^2 \times (5 \times 11111)^2 &= 10^2 \times M^2 \\ 2^2 \times 11111^2 \times 5^2 \times 11111^2 &= 10^2 \times M^2 \\ 10^2 \times 11111^4 &= 10^2 \times M^2 \\ M = 11111^2 &= 123454321. \end{aligned}$$

Qn2. The sum of digits of a 2-digit number cannot exceed $9 + 9 = 18$. Since the sum is 19, the page must have had at least 3 digits.
Since the digit sum of the next page is 2 and the book has less than 1000 pages, this page could be $110, 101$ or 200. Since $110 - 1 = 109$ and $1 + 0 + 9 = 10 \neq 19$, this page is discarded. Similarly page 101 is discarded and therefore this page must be 200 since $200 - 1 = 199$ and $1 + 9 + 9 = 19$ as required. The page that Annet was reading was $200 - 1 = 199$.

Qn3. Since the last two digits of UMS14 and UMS15 sum up to 29, we have that UMS+UMS=1234. Hence UMS=617, so, U=6, M=1, S=7. Thus, U+M+S= $6 + 1 + 7 = 14$.

Qn4. When 2015 is divided by 19, there is a quotient of 106 and a remainder of 1. So the student numbered 1 called out the number 2015.

Qn5. Suppose A lied, then he would be the oldest but in this case all the three would be liars hence, A told the truth and B and C lied. From $C's$ lie, we deduce that A was not the youngest, so, the youngest is either B or C. Since we know that B is lying, then C is not the oldest and therefore he/she must be the youngest.

Qn6. Suppose I have n books altogether, then $n - 2$ of them are algebra, $n - 2$ of them are trigonometry books and $n - 2$ of them are geometry books. Hence $n = 3(n - 2) \implies n = 3$. Therefore I have 3 books altogether.

Qn7.

$$\frac{1 + \frac{3y+y^3}{1+3y^2}}{1 - \frac{3y+y^3}{1+3y^2}} = \frac{\frac{1+3y^2+3y+y^3}{1+3y^2}}{\frac{1+3y^2-3y-y^3}{1+3y^2}} = \frac{1 + 3y^2 + 3y + y^3}{1 + 3y^2 - 3y - y^3} = \frac{(1+y)^3}{(1-y)^3} = \left(\frac{1+y}{1-y}\right)^3.$$

$$H = \log\left(\frac{1+y}{1-y}\right)^3 = 3\log\left(\frac{1+y}{1-y}\right) = 3F.$$

Qn8. Alex's equation is $(x + 2)(2x + 3) = 0 \implies 2x^2 + 3x + 4x + 6 = 0$ or $2x^2 + 7x + 6 = 0$.

Barbra's equation is $(x-2)(x-3)=0 \Longrightarrow x^2-5x+6=0$. From these two equations, $C=6, A=1$ and $B=7$.

The correct equation is $x^2+7x+6=0$. On solving the equation by factorisation method, we get $(x+6)(x+1)=0 \Longrightarrow x=-6$ or $x=-1$ as the solutions to the correct equation.

Qn9. $m=\log_{y^2} x \Longrightarrow x=y^{2m}$ and $n=\log_{x^2} y \Longrightarrow y=x^{2n}$. Thus $x=y^{2m}=(x^{2n})^{2m}=x^{4mn}$. Hence $1=4mn$ so that $n=\frac{1}{4m}$.

But $m+n=1$, so, $m+\frac{1}{4m}=1 \Longrightarrow 4m^2-4m+1=0$. Solving simultaneously gives $m=\frac{1}{2}$. Hence $n=\frac{1}{4\times 0.5}=\frac{1}{2}$. Therefore $x=y$

Qn10. First kilometre : Time $t_1=\frac{Distance}{speed}=\frac{1}{4}$hours.

Second kilometre: Time $t_2=\frac{Distance}{speed}=\frac{1}{x}$hours.

Third kilometre: Time $t_3=\frac{Distance}{speed}=\frac{1}{6}$hours.

$Average\ speed=\frac{total\ distance}{total\ time}=\frac{3}{\frac{1}{4}+\frac{1}{x}+\frac{1}{6}}=\frac{3}{\frac{6x+24+4x}{24x}}=\frac{72x}{24+10x}$

Required is to find an integer value x such that $4<x<60$ and satisfies the equation $\frac{72x}{24+10x}=k$ where k is an integer. This is possible if $24+10x$ is a factor of $72x$. This implies $24+10x$ is a multiple of 72. The possible multiples of 72 are $\{72, 144, 216, 288, \cdots\}$.

Now, $24+10x \neq 72$ since $10x=58$ does not give an integer value of x.

Suppose $x=144$, then $24+10x=144 \Longrightarrow 10x=120$. Hence $x=12$ is the required integer value of x.

SECTION B

Qn11. Let the first term in the geometric progression be $t_1=a$ and r be the common ratio. Then the sequence is $a, ar, ar^2, \cdots ar^{n-1}$. Since $t_1 t_n=3$ then $a.ar^{n-1}=3$ or $a^2r^{n-1}=3$.

$$\begin{aligned} t_1t_2t_3\cdots t_n &= 59049, \quad \text{therefore} \\ (a)(ar)(ar^2)\cdots(ar^{n-1}) &= 59049 \\ a^n r r^2 r^3 r^4 \cdots r^{n-2} r^{n-1} &= 59049 \\ a^n r^{1+2+3+4+\cdots+(n-1)} &= 59049 \\ a^n r^{\frac{1}{2}(n-1)(n)} &= 59049. \end{aligned}$$

From the equation $a^2r^{n-1}=3 \Longrightarrow (a^2r^{n-1})^n=3^n \Longrightarrow a^{2n}r^{(n-1)n}=3^n=(59049)^2$.

Thus $3^n=59049^2 \Longrightarrow n=\frac{2\times\log 59049}{\log 3}=20$. Hence the geometric progression has 20 terms.

Qn12. The last four digits ($GHIJ$) are either 9753 or 7531, and the remaining odd digit (either 1 or 9) is A, B, or C. Since $A+B+C=9$, the odd digit among A, B, and C must be 1. Thus the sum of the two even digits in ABC is 8. The three digits in DEF are

$864; 642$; or 420, leaving the pairs 2 and $0, 8$ and 0, or 8 and 6, respectively, as the two even digits in ABC. Of those, only the pair 8 and 0 has sum 8, so ABC is 810, and the required first digit is 8. The only such telephone number is $810 - 642 - 9753$.

Qn13. Suppose $x \leq 2$, then 2 would be both median and the mode, a contradiction, so $x > 2$.
Suppose $2 < x < 4$, then the list is $2, 2, 2, x, 4, 5, 10$ of which the mode is 2, median is x and mean $\frac{25+x}{7}$. Thus the arithmetic progression is $2, x, \frac{25+x}{7}$. So,

$$x - 2 = \frac{25 - 6x}{7} \Longrightarrow 7x - 14 = 25 - 6x$$

$$13x = 39 \Longrightarrow x = 3.$$

Suppose $x \geq 4$, then the mode is 2, median is 4, and the mean is $\frac{25+x}{7}$ and the arithmetic progression $2, 4, \frac{25+x}{7}$.

$$4 - 2 = \frac{25 + x}{7} - 4 \Longrightarrow 6 = \frac{25 + x}{7} \Longrightarrow x = 42 - 25 = 17.$$

The sum of $x-$values is $3 + 17 = 20$.

Qn14. Since $100e = beh = ceg = def \Longrightarrow h = \frac{100}{b}$, $g = \frac{100}{c}$ $f = \frac{100}{d}$. By comparing rows 1 and 3, then we have $50 \times b \times c = 2 \times \frac{100}{b} \times \frac{100}{c} \Longrightarrow c = \sqrt{\frac{400}{b^2}} = \frac{20}{b} \Longrightarrow b \in \{1, 2, 4, 5, 10, 20\}$.

Thus the product of the entries in the first row is $50 \times b \times \frac{20}{b} = 1000$, so, $100e = 1000 \Longrightarrow e = 10$.
By comparing columns 1 and 3, then we have $50d \times \frac{100}{c} = 2 \times c \times \frac{100}{d}$ from which $d = \frac{c}{5} = \frac{4}{b} \Longrightarrow b \in \{1, 2, 4\}$. It follows from that $c = \frac{1000}{50b} \Longrightarrow c \in \{20, 10, 5.\}$
Now since $g = \frac{1000}{10c}$ then $g \in \{5, 10, 20\}$ and their sum is $5 + 10 + 20 = 35$.

Qn15. We use cosine law, on the triangle $\triangle ABD$ to determine the length of BD.

$$\begin{aligned} BD^2 &= AB^2 + AD^2 - 2(AB)(AD)\cos(< BAD) \\ \cos(< BAD) &= \cos(< BAC + < EAD) \\ &= \cos(< BAC)\cos(< EAD) - \sin(< BAC)\sin(< EAD) \end{aligned}$$

$$= \frac{AC}{AB} \times \frac{AD}{AE} - \frac{BC}{AB} \times \frac{ED}{AE}$$

just because $\triangle ABC$ and $\triangle ADE$ are right angled. But $AB = 75$ and $BC = 21$.
By Pythagoras theorem, $AC = \sqrt{AB^2 - BC^2} = \sqrt{75^2 - 21^2} = 72$.
Now, $AC = 72, CE = 47 \Longrightarrow AE = AC - CE = 25$.
Since $AE = 25, AD = 20 \Longrightarrow ED = \sqrt{AE^2 - AD^2} = \sqrt{25^2 - 20^2} = \sqrt{225} = 15$.

$\cos(< BAD) = \frac{AC}{AB} \times \frac{AD}{AE} - \frac{BC}{AB} \times \frac{ED}{AE} = \frac{72 \times 20}{75 \times 25} - \frac{21 \times 15}{75 \times 25} = \frac{3}{5}$.

$BD = \sqrt{75^2 + 20^2 - (2 \times 75 \times 20 \times \frac{3}{5})} = \sqrt{4225} = 65$. Therefore $BD = 65$.